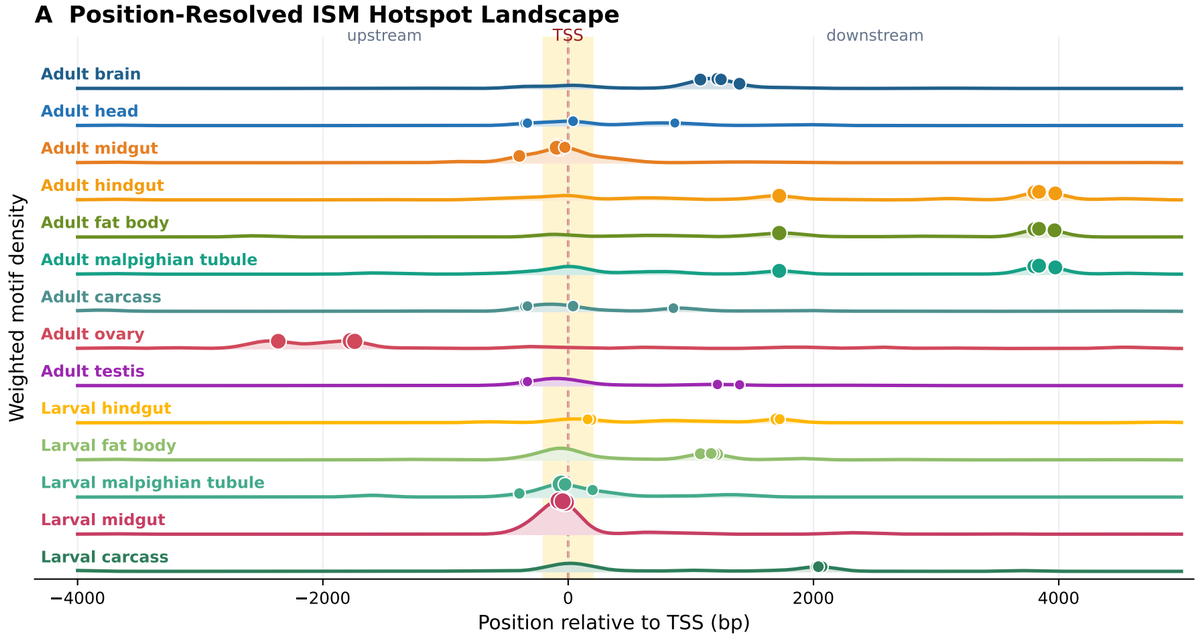
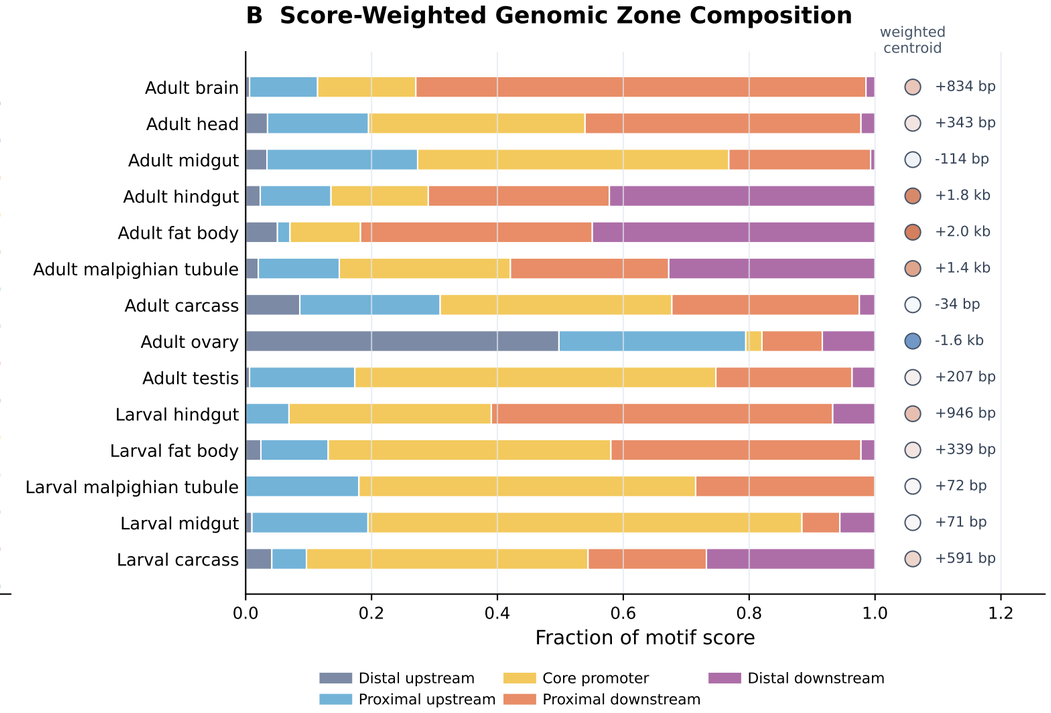
ISM interpretability summary across 14 tissues

Panel A shows score-weighted hotspot landscapes, Panel B summarizes regional allocation, Panel C highlights recurrent exact 12-mers, and Panel D selects one tissue-enriched exact 12-mer per tissue.

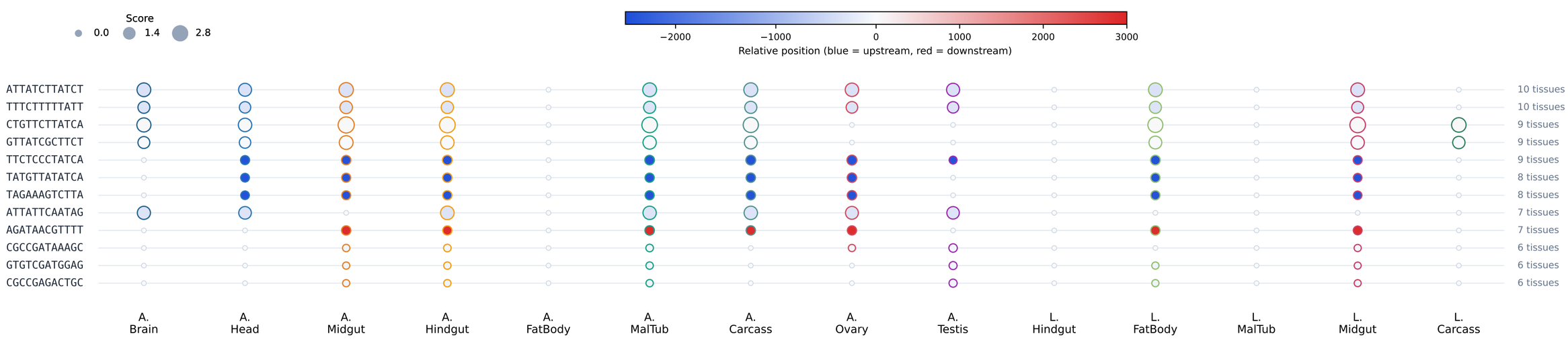
A Position-Resolved ISM Hotspot Landscape



B Score-Weighted Genomic Zone Composition



C Cross-Tissue Shared Motif Atlas



D Tissue-Enriched Exact-Motif Signatures

Adult brain
Signature exact 12-mer
AATGAAAGTTAA
+1.1 kb | Proximal downstream
ISM 3.52 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Adult head
Signature exact 12-mer
GGTGATAAGATT
+870 bp | Proximal downstream
ISM 1.17 | seen in 2/14 tissues
next: A.Carcass 1.04
14-tissue footprint A | L

Adult midgut
Signature exact 12-mer
TGATAAAGAATC
-398 bp | Proximal upstream
ISM 3.44 | seen in 2/14 tissues
next: L.MalTub 2.29
14-tissue footprint A | L

Adult hindgut
Signature exact 12-mer
TCTTATCTTATC
+3.1 kb | Distal downstream
ISM 2.00 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Adult fat body
Signature exact 12-mer
ATATAATAACGC
+4.0 kb | Distal downstream
ISM 5.60 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Adult malpighian tubule
Signature exact 12-mer
TAACGCATATGC
+4.0 kb | Distal downstream
ISM 5.64 | seen in 2/14 tissues
next: A.Hindgut 5.54
14-tissue footprint A | L

Adult carcass
Signature exact 12-mer
TATTAAGATAAC
+858 bp | Proximal downstream
ISM 1.73 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Adult ovary
Signature exact 12-mer
ATTATCAATATT
-1.8 kb | Proximal upstream
ISM 8.30 | seen in 2/14 tissues
next: L.Carcass 3.00
14-tissue footprint A | L

Adult testis
Signature exact 12-mer
TGATATCAGATA
+1.2 kb | Proximal downstream
ISM 1.46 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Larval hindgut
Signature exact 12-mer
TGATAAGAAAGA
+1.7 kb | Proximal downstream
ISM 3.11 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Larval fat body
Signature exact 12-mer
TTTGTCGAGACA
+1.2 kb | Proximal downstream
ISM 3.56 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Larval malpighian tubule
Signature exact 12-mer
GTTATCTTATCT
-56 bp | Core promoter
ISM 9.56 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Larval midgut
Signature exact 12-mer
TGATAAGGAGTG
-33 bp | Core promoter
ISM 10.83 | seen in 1/14 tissues
next: none
14-tissue footprint A | L

Larval carcass
Signature exact 12-mer
AACGTTATCTCT
+2.1 kb | Distal downstream
ISM 2.74 | seen in 1/14 tissues
next: none
14-tissue footprint A | L